

Chutong Yang

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Research Focus

My research focuses on learning theory and algorithms for reliable machine learning, with an emphasis on calibration and optimization. My recent work explores LLM reasoning and multi-agent collaboration, including the design and evaluation of AI agents for automated research.

Education

The University of Texas at Austin , Ph.D. in Computer Science • Recipient, Amazon AI PhD Fellowship	Aug 2023 – Present
Stanford University , M.S. in Computer Science • Full financial support for two years.	Sep 2021 – Jun 2023
University of California, San Diego , B.S. in Computer Science and Mathematics	Sep 2016 – Jun 2020

Experience

Amazon Web Services (AWS) , Applied Scientist Intern • Built TCSAlgBench, a benchmark of sandboxed, research-level theoretical computer science challenges for evaluating LLMs and agentic systems. • Designed and evaluated agent harnesses, including plan-and-execute and Monte Carlo tree search (MCTS), finding that agentic planning performed best among the evaluated approaches.	Summer 2026
The University of Texas at Austin , Research Assistant • Developed optimal online recalibration algorithms, with simultaneous calibration and squared-loss improvement guarantees for discretized forecasts under adversarial data sequences. • Develop methods for reliable LLM reasoning through uncertainty-guided multi-agent deliberation and RL-based confidence calibration.	Aug 2023 – Present
Stanford University , Research Assistant • Researched algorithmic fairness with Prof. Omer Reingold, focusing on multicalibration. • Developed optimization algorithms with Prof. Aaron Sidford using interior-point methods.	2022 – 2023

Skills

Programming: Python, C++, Java, MATLAB **ML:** PyTorch

Publications

(Authors are listed alphabetically unless otherwise indicated by *.)

13. TCSAlgBench: Benchmarking Automated Proving for Research-Level Theoretical Computer Science <i>Chutong Yang*</i> , Xiyuan Zhang, Yu Huang, Boran Han, Soonho Kong, Shuai Zhang, Vihang Prakash Patil, Zhen Han, Michael Bohlke-Schneider, Bernie Wang http://arxiv.org/abs/2609.35606	Preprint, 2026
12. Gap-Free Streaming PCA Beyond Rank-One Updates: Near-Optimal Rates and Applications to Differential Privacy Anming Gu, Syamantak Kumar, Kevin Tian, <i>Chutong Yang</i> https://arxiv.org/abs/2609.26508	Preprint, 2026
11. Optimal Recalibration of an Online Predictor Lunjia Hu, Kevin Tian, <i>Chutong Yang</i> https://arxiv.org/abs/2607.19689	NeurIPS 2026 (spotlight)

- 10. Simultaneous Blackwell Approachability and Applications to Multiclass Omniprediction** COLT 2026
Lunjia Hu, Kevin Tian, *Chutong Yang*
<https://arxiv.org/abs/2602.17577>
- 9. Spherical Leech Quantization for Visual Tokenization and Generation** CVPR 2026 (Highlight)
Yue Zhao*, Hanwen Jiang, Zhenlin Xu, *Chutong Yang*, Ehsan Adeli, Philipp Krähenbühl
<https://arxiv.org/abs/2512.14697>
- 8. Proportionality from Low-Dimensional Approval Data** AAMAS 2026
Zhiyi Huang, Gregory Kehne, *Chutong Yang*
Extended abstract: <https://doi.org/10.65109/YHKI7695>
- 7. Private Geometric Median in Nearly-Linear Time** NeurIPS 2025
Syamantak Kumar, Daogao Liu, Kevin Tian, *Chutong Yang*
TPDP 2025 Workshop
<https://arxiv.org/abs/2505.20189>
- 6. Omnipredicting Single-Index Models with Multi-Index Models** STOC 2025
Lunjia Hu, Kevin Tian, *Chutong Yang*
<https://arxiv.org/abs/2411.13083>
- 5. Testing Calibration in Nearly-Linear Time** NeurIPS 2024
Lunjia Hu, Arun Jambulapati, Kevin Tian, *Chutong Yang*
<https://arxiv.org/abs/2402.13187>
- 4. Omnipredictors for Constrained Optimization** ICML 2023
Lunjia Hu, Inbal Livni-Navon, Omer Reingold, *Chutong Yang*
<https://arxiv.org/abs/2209.07463>
- 3. Active Learning Polynomial Threshold Functions** NeurIPS 2022
Omri Ben-Eliezer, Max Hopkins, *Chutong Yang*, Hantao Yu
<https://arxiv.org/abs/2201.09433>
- 2. A Fast Exact Algorithm for Deployment of Sensor Nodes for Internet of Things** ISF 2020
Qinghua Zheng*, *Chutong Yang*, Haijun Yang, Jianhe Zhou
Information Systems Frontiers: <https://link.springer.com/article/10.1007/s10796-018-9890-3>
- 1. Detailed placement for IR drop mitigation by power staple insertion in sub-10nm VLSI** DATE 2019
Sun ik Heo, Andrew B Kahng, Minsoo Kim, Lutong Wang, *Chutong Yang*
<https://ieeexplore.ieee.org/abstract/document/8715096>

Professional Service

Conference reviewer: NeurIPS 2024, ICLR 2025, ICML 2025, ITCS 2026